MILITARY SPECIFICATION

IGNITION ELEMENT, ELECTRICAL, PARTS FOR

1. SCOPE

1.1 This specification covers all parts for an electrical ignition element intended for use with 152MM Cartridges.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons; General Specification for.
MIL-I-45607		Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-1168	-	Lot Numbering of Ammunition.
MIL-STD-1167	-	Ammunition Data Cards.
MIL-STD-286	-	Propellants, Solid: Sampling, Examinations and Testing
MIL-STD-414	-	Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

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DRAWINGS

ORDNANCE

9249027	Front Support Assembly
9249026	Subassembly

PUBLICATIONS

ORDNANCE CORPS

ET-9249026	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM D-792	Specific Gravity and Density of Plastics by Displacement.
ASTM-D-1621	Compressive Strengh of Rigid Cellular Plastics

(Copies may be purchased directly from the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103).

DEPARTMENT OF TRANSPORATION CODE OF FEDERAL REGULATIONS, TITLE 49, Parts 0-190.
(The Department of Transporation Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, D.C , 20402. Orders should cite "49 CFR 0-190 (Latest Revision).)"

3 REQUIREMENTS

3.1 Material.-The materials and parts for the sub-assembly shall be in accordance with the applicable drawings and specifications listed in 2.1. The materials for the outer cup and the inner sleeve shall have a tolerance of ± 1 percent for tin, ± 0.75 percent for the zinc, and ± 0.5 percent for cadmium.

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3.2 Stability.-The propellant for the front support shall not completely change the color of the methyl voilet paper to salmon pink color in less than sixty (60) minutes when tested in accordance with 4.3.3.1.

3.3 Bulk Density.-The bulk density of the front support shall be 1.30 grams per cubic centimeter maximum when tested as specified in 4.3.3.2.

3.4 Relative Quickness.-The relative quickness of the front support shall be not less than 300 percent, when compared to M26E1 Propellant Lot RAD-66589 or equivalent, and tested in accordance with 4.3.3.3 (see 6.5).

3.5 Crush strength.-The crush strength of the front support shall be 100 pounds minimum in the vertical direction and 40 pounds minimum in the horizontal direction when tested as specified in 4.3.3.4.

3.6 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of chips, cracks, dirt, grease, flux, tears, freyed edges, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.7 Permanent set.-The permanent set of the spring shall be 3.0 percent maximum when tested as specified in 4.3.3.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission -Prior to initiation of regular production, the contractor shall submit a first article sample to the Government facility designated by the contracting officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 25 bridge ribbons, 25 inner sleeves, 25 outer cups, 150 front supports, 25 cushions, 25 springs, 25 collars, 25 discs and 25 buttons which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment, which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be accompanied by certificates of analysis for materials. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply so as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defectives will not be submitted unless specifically authorized by the contracting officer.

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4.2.2 Inspections to be performed.-Parts will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 **and** 4.3.3 of this specification, and any or all requirements of the applicable drawings. All tests shall be performed as specified in 4.4.

4.2.3 Rejection.-If any component fails to comply with the requirements for the inspections performed in accordance with 4.2.1 , the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any or all of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required shall be in accordance with MIL-STD-1168.

4.3.1.1 Parts.-A lot of inner sleeves, outer cups, cushions, springs, collars, discs, buttons, or front supports shall consist of parts produced by one manufacturer in one unchanged process in accordance with the same drawing, and same drawing revision, same specification and same specification revision. Drawing, specification, and process changes not affecting safety, performance or interchangeability, as determined by the Government shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of inspection lot interfix numbers shall be in accordance with Standard MIL-STD-1168.

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4.3.1.2 Bridge ribbons.-A lot shall consist of all bridge ribbons produced by one manufacturer to the same drawing, same drawing revision, same specification, and same specification revision using the same punch and die and stamped from one continuous roll of preformed ribbon which has been certified by the supplier as complying with the applicable material standard. The criteria and procedures for the assignment of inspection lot interfix numbers shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.5).

4.3.2.1 Cup, outer (see drawing 9204178 covering a detail of drawing 9249026).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Diameter, inner	Gage	01001
102.	Thickness	Gage	01002
103.	Length	Gage	01003
104.	True position of lance pattern diameter to I.D. of cup	Gage	01004
Minor:	AQL 1.5 percent		
201.	Diameter, hole	Gage	01005
202.	Location of hole	Gage	01006
203.	Evidence of poor workmanship (see 3.6)	Visual	01007
204.	Cut out extends completely through hole	Visual	01008

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4.3.2.2 Sleeve, inner (see drawing 9247426 covering a detail of drawing 9249027).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Outer diameter of tube	Gage	02001
102	Thickness	Gage	02002
103.	Total length	Gage	02003
Minor:	AQL 1.00 percent		
201.	Diameter of flange	Gage	02004
202.	Evidence of poor workmanship (see 3.6)	Visual	02005

4.3.2.3 Support, front (see drawing 9204184 covering a detail of drawing 9249027).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	1.0 percent		
101.	Outer diameter, min	Gage	03001
102.	Diameter of sleeve support hole, max	Gage	03002
103.	Diameter of plunger hole, min	Gage	03003
104.	Depth of plunger hole	Gage	03004
105.	Concentricity of plunger hole to outside diameter	Gage	03005
106.	Concentricity of sleeve support hole to outside diameter	Gage	03006
Minor:	AQL 2.5 percent		
201.	Overall length	Gage	03007
202.	Depth of ignition charge well	Gage	03008
203	Width of ignition charge well, min	Gage	03009
204.	Presence of test hole	Visual	03010
205.	Evidence of poor workmanship (see 3.6)	Visual	03011

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4.3.2 4 Bridge ribbon (see drawing 9204181 covering a detail of drawing 9249027).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Distance between radii along width.	Gage	04001
102.	Thickness of ribbon.....	Gage	04002
Minor:	AQL 1.00 percent.		
201.	Overall length less than minimum	Gage	04003
202.	Evidence of poor workmanship (see 3.6).....	Visual	04004

4.3.2.5 Cushion (see drawing 9247424 covering a detail of drawing 9249027).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101	Outside diameter.....	Gage	05001
102.	Inside diameter.....	Gage	05002
103	Length....	Gage	05003
104.	Concentricity and perpendicularity, max.	Gage	05004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6).....	Visual	05005

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4 3.2.6 Spring (see drawing 9248753 covering a detail of drawing 9249027).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Free length	Gage	06001
102	Outside diameter	Gage	06002
103.	Solid height (max)	Gage	06003
104.	Number of coils	Visual	06004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	06005

4.3.2.7 Collar (see drawing 9247425 covering a detail of drawing 9249027).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Small diameter	Gage	07001
102.	Inside diameter of outer edge	Gage	07002
103.	Overall length	Gage	07003
104.	Thickness	Gage	07004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	07005

4.3 2.8 Disc (see drawing 9247422 covering a detail of drawing 9249027).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Inside diameter	Gage	08001
102	Outside diameter	Gage	08002
103.	Thickness	Gage	08003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	08004

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4.3.2.9 Button (see drawing 9247427 covering a detail of drawing 9249027).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Outside diameter	Gage	09001
102.	Thickness	Gage	09002
103.	Spherical radius	Gage	09003
Minor:	AQL 1.00 percent		
201.	Height of lip.	Gage	09004
202.	Evidence of poor workmanship (see 3.6)	Visual	09005

4.3.3 Testing

4.3.3.1 Stability (see 3.2), Major defect, Code No. 10001.-A stability heat test at 120 degrees C shall be performed in accordance with MIL-STD-286, Method 404.1.2. A random sample of five test specimens of 2.5 grams each shall be tested for each lot of propellant. Failure of any sample to meet the requirement of 3.2 is cause for lot rejection.

4.3 3.2 Bulk Density (see 3.3), Major defect, Code No. 11001.-Lot sampling shall be in accordance with MIL-STD-414, Inspection Level IV with an AQL of 0.40 percent. Bulk density of individual front supports shall be determined in accordance with ASTM-D-792 using water as the immersion liquid. If the sample fails the acceptance criteria, the lot shall be rejected (see 6.6).

4.3.3.3 Relative Quickness (see 3.4), Major defect, Code No. 12001.-The relative quickness of the front support shall be determined at 8500 psi using a sufficient number of randomly selected front supports per lot to provide a loading density of 0.2g/cc in a nominal 200 cc closed bomb. A minimum of three separate tests shall be conducted for each lot. Testing shall be in accordance with MIL-STD-286, Method 801.1 and shall be conducted at 70 $\pm$ 2°F. If the average of the three values obtained fails the requirement of 3.4, the lot of front supports shall be rejected.

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4.3.3.4 Crush strength (see 3.5), Major defect, Horizontal direction Code No. 13001, Vertical direction Code No. 13002.- Lot sampling for each defect shall be in accordance with MIL-STD-414, Inspection level IV with an AQL of 0.40 percent. Front support, crush strength in each direction shall be determined in accordance with ASTM-D-1621. If either sample fails the acceptance criteria, the lot shall be rejected (see 6.6).

4.3.3.5 Permanent set (see 3.7), Major defect, Code No. 14001.-Lot sampling shall be in accordance with MIL-STD-414, Inspection Level IV with an AQL of 0.40 percent. Permanent set of individual springs shall be determined as specified in 4.4.1. If the sample fails the acceptance criteria, the lot shall be rejected (see 6.6).

4.3.4 Inspection Equipment.-Equipment Tabulation Number ET-9249026 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4 2 Government use of contractor's inspection and test equipment'-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

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4.4.1 Permanent set.-Impart a load to each spring at a rate of 9 inches per second until the spring is compressed to a length of 0.190 ± .010 inch. When the spring is compressed, release the load and allow the spring to return to its uncompressed free length. Repeat the cycle an additional eight times. At the conclusion of the final cycle, measure the free length of the spring (length B). Calculate percent permanent set for each spring using the following equation:

$$\% \text{ Permanent set} = \frac{A - B}{A} \times 100$$

where:
A = Free length of spring prior to test
B = Free length of spring after nine compression cycles.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking

5.1.1 Level C.-Inert Parts shall be packaged, packed and marked in accordance with Military Standard 1169. Explosive components shall be packaged, packed and marked in accordance with Department of Transportation Code of Federal Regulations, Title 49, Parts 0-190.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:
a. Title, number and date of this specification.
b Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the Inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

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6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU) and non-mandatory (indicated bydrawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commerical equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in

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accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format normally employed by the contractor provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional marking of clearly understandable, high contrast reproduction. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND12. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submssion of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of final submission of designs.

6.4 Submission of first article sample.-Instructions as to the location for the evaluation of first article samples shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquires should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NK1, Dover, New Jersey 07801.

6.5 Relative Quickness.-Comparison shall be made with M26E1 propellant to assure satisfactory ignition delay.

6.6 MIL-STD-414.-Estimates of lot percent defective for bulk density, horizontal and vertical crush strength and permanent set shall be calculated as in Example B-2.

Custodian:
Army-MU

Preparing Activity:
Army-MU

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